

BABU BANARASI DAS **UNIVERSITY**



SCHOOL OF COMPUTER APPLICATIONS

ASSIGNMENT ON
DESCRIPTIVE ANALYTICS
(BCADSN23201)

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Descriptive Analytics Report: **Daily Purchase Behaviour at Scale**

Tracking Best-Sellers, Regional Demand, Order Status Distribution, Category wise sales, Top cities by sales, Quantity sold by category, Quantity sold by product size

Note on data: This report uses an Amazon India sales dataset containing **70 order records** across **4 product categories, 11 states and 16 cities**, covering orders placed on **April 30, 2022**. The dataset includes order status, quantity, sales amount, delivery method, customer location, shipping status, promotions and B2B information. It is designed to support analysis of **product performance, sales, regional demand and category trends**.

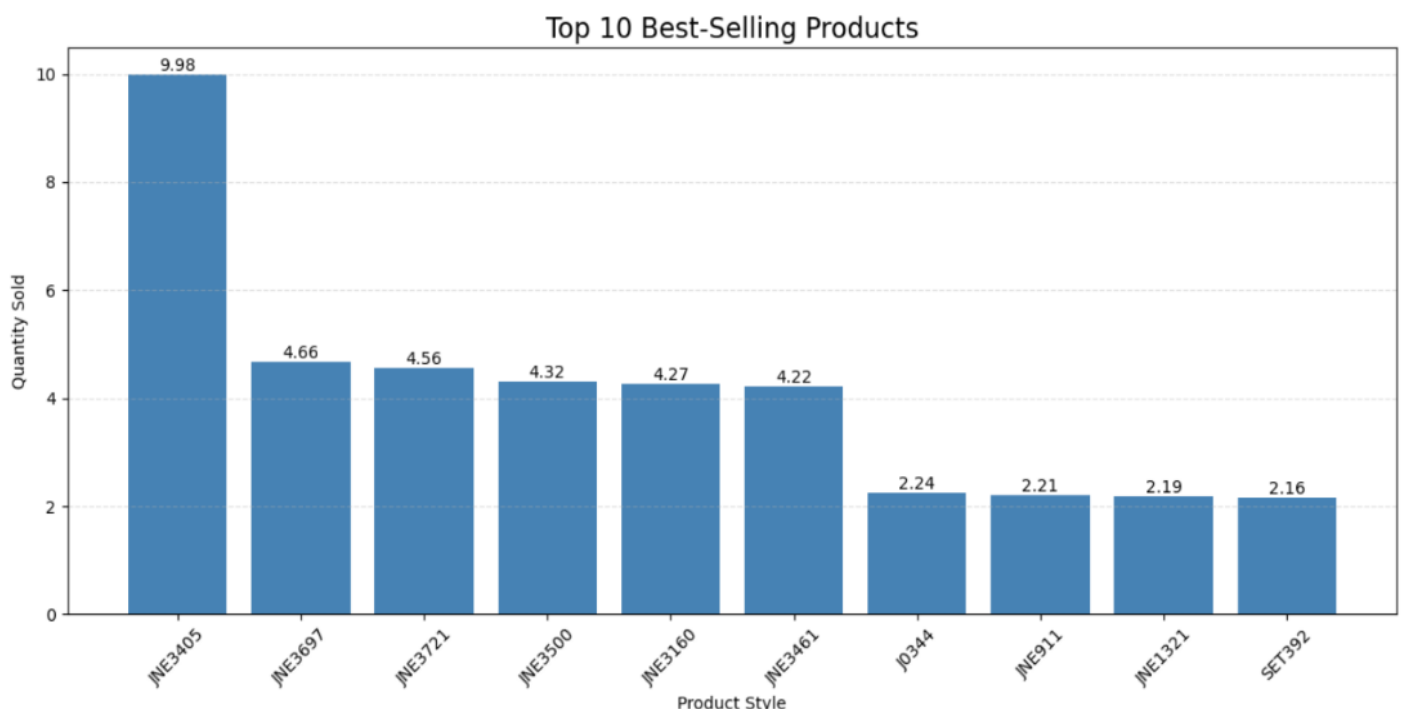
2. Objectives

- I. What are the top 10 best-selling products based on quantity sold?
- II. What is regional sales demand and what are the top-performing states?
- III. What is the distribution of order statuses within the dataset?
- IV. What is category-wise sales revenue and what are the major revenue-generating categories?
- V. What are the top 10 cities by sales revenue?
- VI. What is the quantity sold across product categories and understand product demand?
- VII. What is the contribution of different fulfilment methods to total sales revenue?
- VIII. What are popular product sizes and what is their contribution to sales performance?

4. How this would work at real scale?

LAYER	PURPOSE	TYPICAL TOOLS
Data Collection	Store sales transaction records including product category, revenue, ratings, discounts, and customer information	CSV Dataset
Data Loading	Import and load the dataset into the analysis environment	Pandas
Data Cleaning	Check missing values, data types, duplicates, and prepare data for analysis	Pandas
Data Analysis	Calculate revenue, ratings, regional performance, payment method usage, and product performance metrics	Pandas, NumPy
Data Visualization	Create charts and graphs to identify trends and patterns	Matplotlib
Reporting	Present findings and business insights in a structured notebook report	Jupyter Notebook

5. Top 10 Best-Selling Products



The **Top 10 Best-Selling Products** graph compares products according to the quantity recorded in the selected Amazon sales sample. The leading product in the analysis records approximately **9.98 units**, representing **10.46% of the total recorded quantity of 95.45 units**. The leading product therefore contributes a noticeably larger share of volume than the other individual products in the sample. The strongest product-level performance is associated with the **Kurta category**, which is also the overall leading category by quantity. The graph shows that **JNE3405** is the **leading product**, with approximately **9.98 units** recorded. It is followed **by JNE3697** with approximately **4.66 units** and **JNE3721** with approximately **4.56 units**. Other products appearing in the top ten include **JNE3500** with approximately **4.32 units**. All four of them belongs to the category of **Kurta**. **JNE3160** with approximately **4.27 units**, and **JNE3461** with approximately **4.22 units** and they belong to the category of **Set** and **Western Dress** respectively.

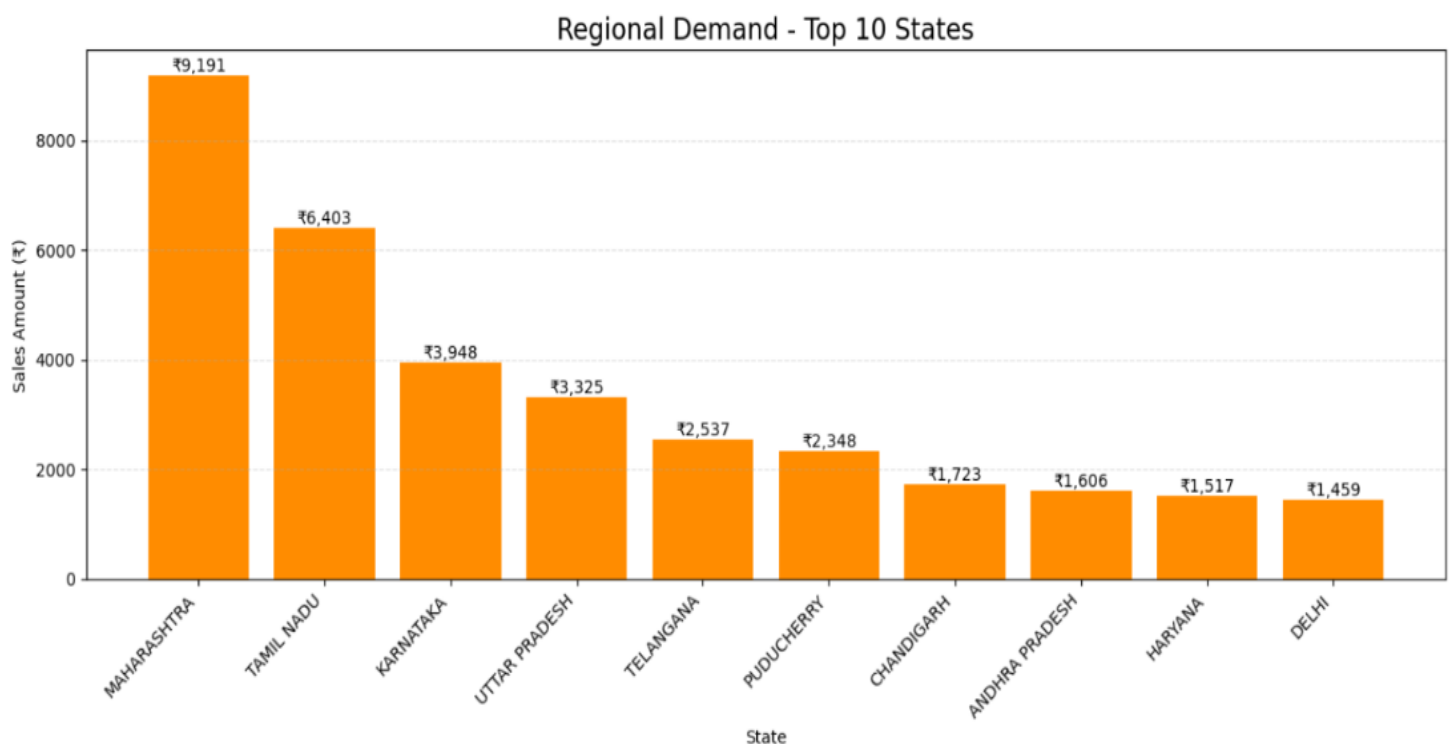
JNE3405 therefore stands out considerably from the other products in the extracted sample. Its recorded quantity is more than twice that of JNE3697, the second-ranked product. This indicates that within this particular sample, JNE3405 has the strongest recorded product-level demand.

- **KPI**

Top-selling product: JNE3405 – Category: Kurta

Recorded quantity: 9.98 units

6. Regional Demand



The **Regional Demand** graph shows that **Maharashtra is the leading state**, recording approximately **₹9,191.04 in sales** and **24.28 units of quantity**. Maharashtra contributes approximately **25.93% of total recorded sales**, making it the strongest geographical market in the selected sample. **Tamil Nadu** follows with approximately **₹6,402.85**, representing **18.06%** of total sales, while **Karnataka** ranks third with approximately **₹3,948.21**, contributing **11.14%**.

Together, these three states account for approximately **55.13% of the total recorded sales**, indicating that more than half of the sample's sales value is concentrated in these leading markets. This concentration can be useful for understanding regional demand and planning distribution, inventory allocation, and fulfilment resources.

The least regional demand is from **Delhi**.

Maharashtra therefore contributes approximately **25.9% of the total recorded sales amount**, making it the most important state in this particular extract. The difference between Maharashtra and Tamil Nadu is approximately **₹2,788**, demonstrating a noticeable gap between the two leading states.

The results show that sales are concentrated in a limited number of geographical regions. Such information can be useful for understanding **regional demand, logistics planning, distribution, and location-based marketing strategies**.

Because this is a small extracted dataset, the results should be interpreted as the geographical pattern of the selected records rather than the overall geographical performance of Amazon.

- **KPI**

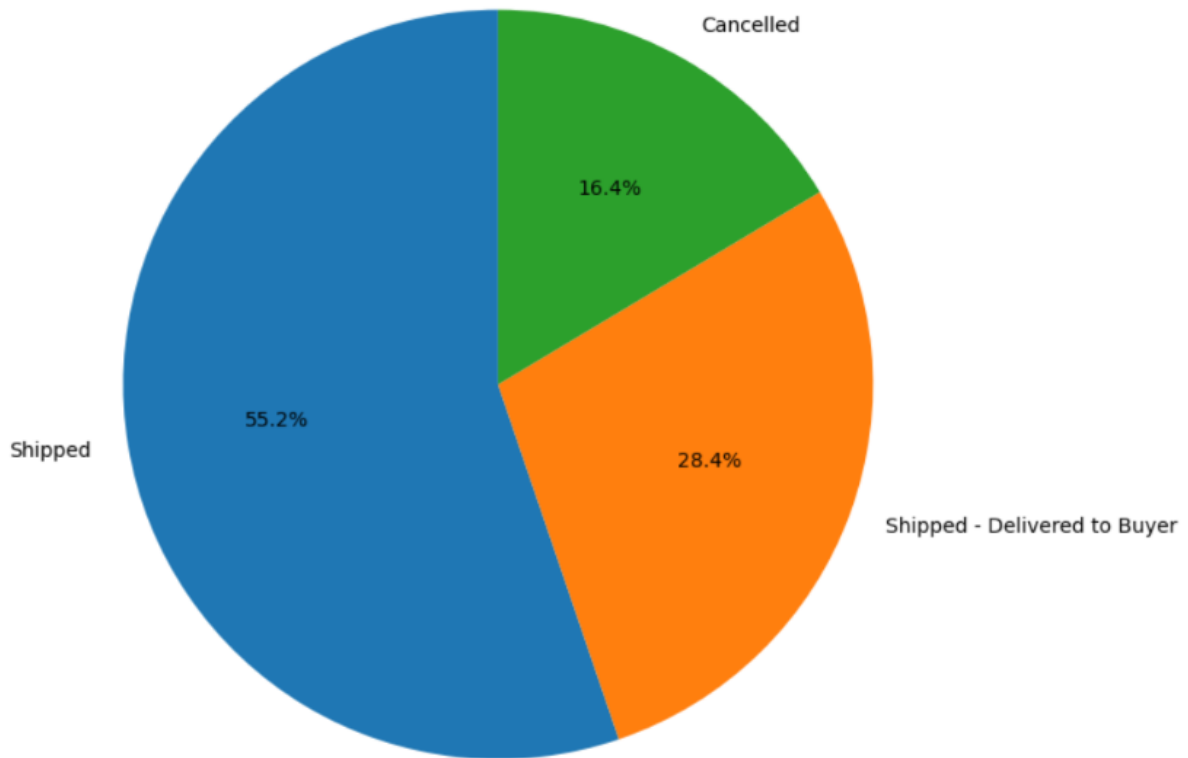
Top-performing state: Maharashtra

Recorded sales: ₹9,191.04

Share of total sales: ≈25.9%

7. Order Status Distribution

Order Status Distribution



The **Order Status Distribution** shows that **Shipped** is the most common order status, with **37 out of 67 orders (55.22%)**. When the shipped orders are further analysed by state, **Tamil Nadu has the highest number of shipped orders, with 12 orders**, representing **32.43% of all shipped orders**. It is followed by **Maharashtra with 7 orders (18.92%)** and **Uttar Pradesh with 6 orders (16.22%)**.

Among the states having shipped orders, **Karnataka records the lowest number with 3 shipped orders**, representing **8.11% of all shipped orders**.

Chandigarh, Delhi, and Rajasthan also record 3 shipped orders each.

This regional breakdown shows that **Tamil Nadu is the strongest region for the Shipped status**, while Karnataka has the lowest shipped-order count among the states where shipped orders are recorded. The concentration of shipped orders in Tamil Nadu indicates relatively stronger order fulfilment activity in this state within the selected sample.

KPI

Highest Shipped Region: Tamil Nadu — 12 orders (32.43%)

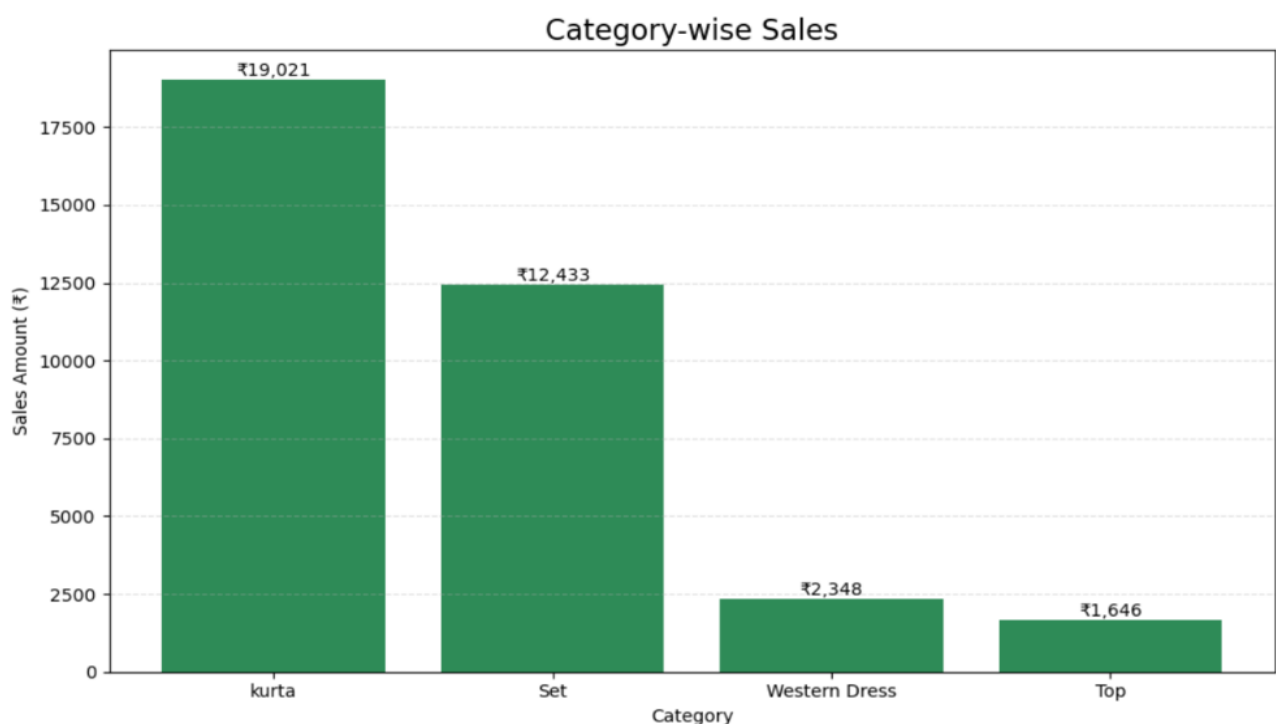
Lowest Shipped Region: Karnataka — 3 orders (8.11%)

Additional Status-Level Insight

The same regional analysis can also be applied to the other two order statuses:

- **Shipped – Delivered to Buyer:** Maharashtra leads with **6 orders (31.58%)**, while Haryana has the lowest with **3 orders (15.79%)** among states with this status.
- **Cancelled:** Maharashtra has the highest number with **4 cancelled orders (36.36%)**, followed by Puducherry with **4 orders (36.36%)**. Telangana records **3 cancelled orders (27.27%)**.

8. Category-wise Sales



The **Category-wise Sales** analysis evaluates the amount of revenue generated by each product category. The four categories represented in the extracted dataset are **Kurta, Set, Western Dress, and Top**.

The results show that **Kurta is the highest revenue-generating category, with approximately ₹19,020.51 in recorded sales**. Set ranks second with approximately ₹12,433.22, while Western Dress generates approximately ₹2,348.45. Top records approximately ₹1,646.23.

Kurta contributes approximately **53.7% of the total recorded sales amount**, making it the dominant category in this dataset. The difference between Kurta and Set is approximately ₹6,587, indicating that Kurta generates substantially more revenue within this sample.

The analysis demonstrates that revenue is concentrated in the two major categories, particularly Kurta and Set. This information can help in understanding **category performance, product planning, promotional focus, and revenue contribution.**

- **KPI**

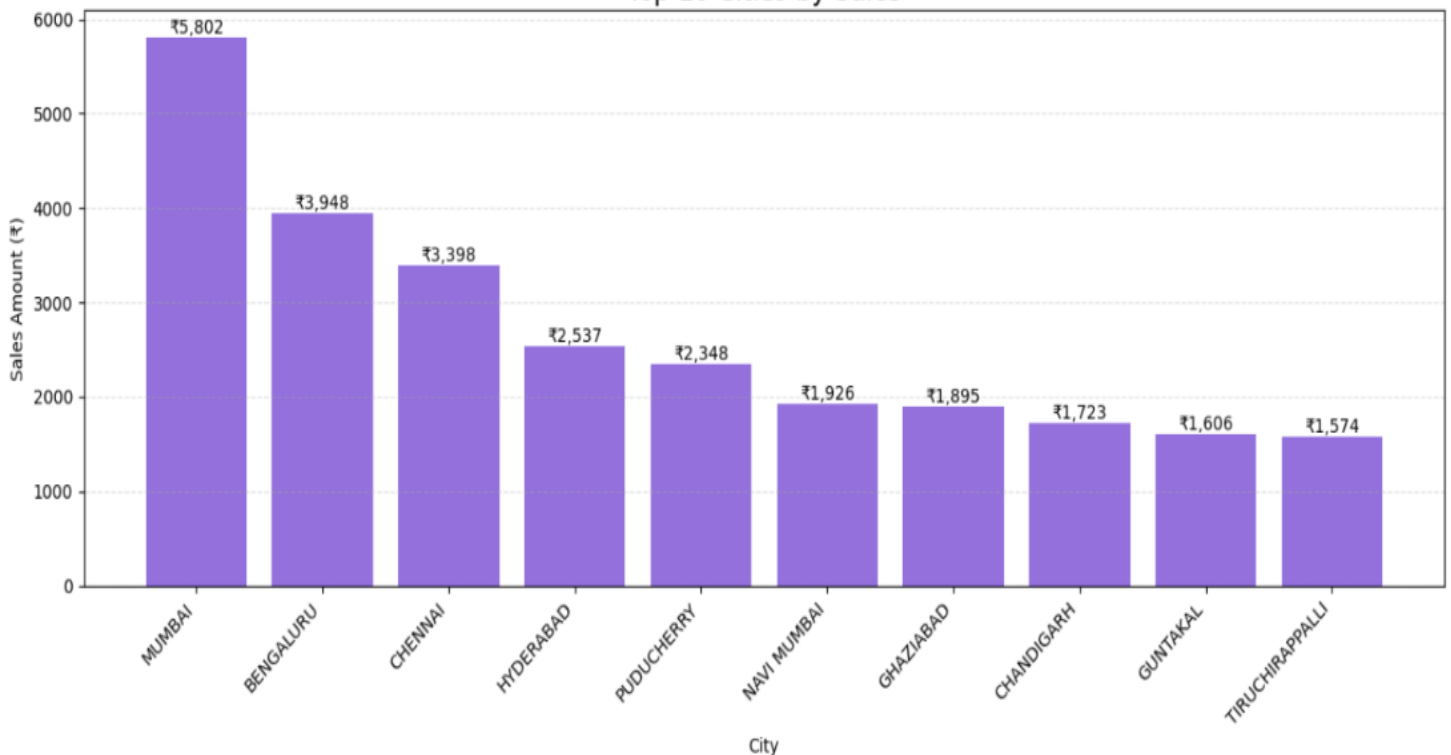
Highest revenue category: Kurta

Recorded sales: ₹19,020.51

Revenue contribution: ≈53.7%

9. Top 10 Cities by Sales

Top 10 Cities by Sales



The **Top 10 Cities by Sales** analysis examines sales concentration at the city level. It uses the shipping city and recorded sales amount to identify the cities contributing the highest sales within the selected dataset.

Mumbai is the leading city, generating approximately **₹5,801.73**, which represents **16.37% of total recorded sales.**

Bengaluru follows with approximately **₹3,948.21** or **11.14%**, while **Chennai** contributes approximately **₹3,397.88** or **9.59%**. **Hyderabad** ranks next with approximately **₹2,536.96** or **7.16%**.

The four leading cities together account for approximately **44.25% of the total recorded sales**, showing that a significant share of the sample's sales is concentrated in a relatively small number of cities. This information can

support decisions related to regional inventory allocation, delivery planning, and targeted promotions.

KPI: Leading city = **Mumbai — ₹5,801.73 (16.37% of total sales).**

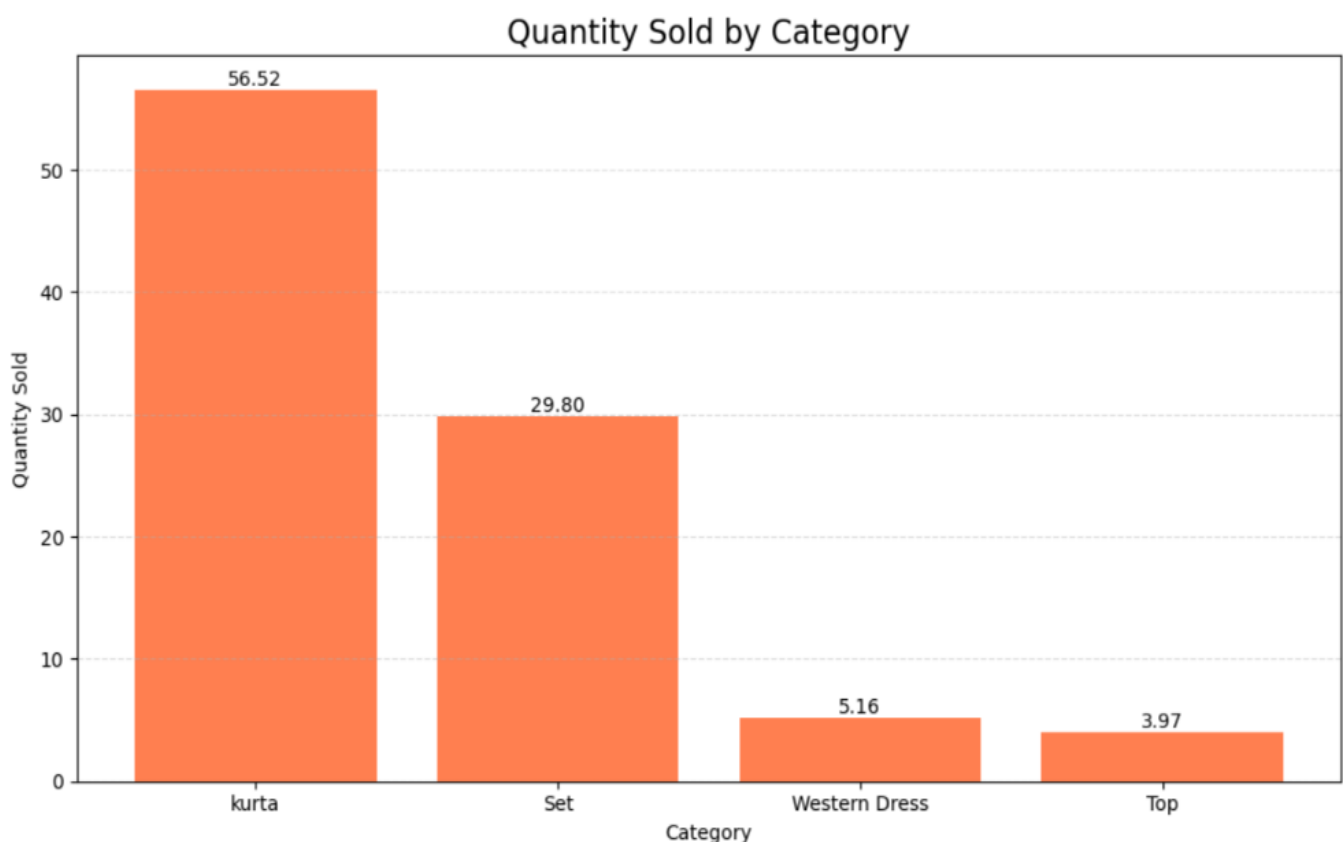
KPI

Top-performing city: Mumbai

Recorded sales: ₹5,801.73

Share of total sales: ≈16.4%

10. Quantity Sold by Category



The **Quantity Sold by Category** analysis measures the total recorded quantity associated with each product category. Unlike Category-wise Sales, which measures revenue, this analysis focuses on **product volume**.

The results show that **Kurta has the highest recorded quantity at approximately 56.52 units**, followed by **Set with 29.80 units**. Western Dress records approximately **5.16 units**, while Top records approximately **3.97 units**. Kurta therefore accounts for approximately **59.2% of the total recorded quantity**, making it the most significant category in terms of sales volume. The difference between Kurta and Set is approximately **26.72 units**, showing a substantial gap in recorded quantity.

This analysis complements Category-wise Sales. In this dataset, Kurta is the leading category in both **revenue and quantity**, which makes it particularly important in the selected sample.

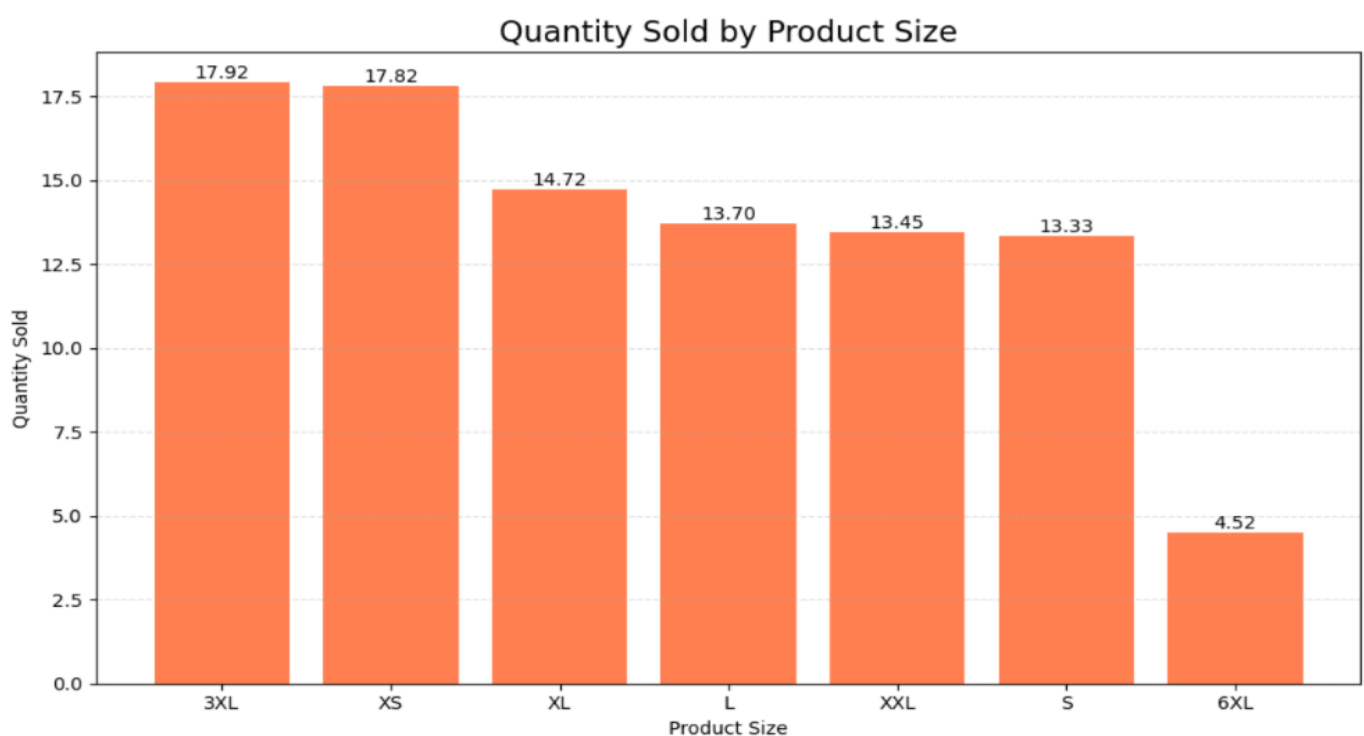
- **KPI**

Highest-volume category: Kurta

Recorded quantity: 56.52 units

Share of total quantity: ≈59.2%

13. QUANTITY SOLD BY PRODUCT SIZE



The **Quantity Sold by Product Size** analysis examines the distribution of recorded product quantities across different apparel sizes. Since the dataset contains fashion-related categories such as Kurta, Set, Western Dress, and Top, size-level analysis provides additional information about the demand pattern within the sample.

The results show that **3XL has the highest recorded quantity at approximately 17.92 units**, followed closely by **XS with approximately 17.82 units**. XL records approximately **14.72 units**, while L records **13.70 units** and XXL records approximately **13.45 units**.

The difference between the highest and second-highest sizes is relatively small, at approximately **0.10 units**, indicating that 3XL and XS have almost identical recorded quantities in this particular extract.

The analysis can help demonstrate how descriptive analytics can be used for **size-level demand analysis and inventory planning**. However, because this dataset is a small sample, these values should not be treated as a representation of overall Amazon customer size preferences.

- **KPI**

Highest-demand size: 3XL

Recorded quantity: 17.92 units

14. Summary of Key Metrics

METRIC	VALUE
Usable sales records	67
Total recorded quantity	95.45 units
Total recorded sales	₹35,448.40
Average sales amount	₹537.10
Top-selling product	JNE3405(Kurta) — 9.98 units
Top state by sales	Maharashtra — ₹9,191.04
Top city by sales	Mumbai — ₹5,801.73
Top revenue category	Kurta — ₹19,020.51
Most common order status	Shipped — 37 records
Highest quantity by size	3XL — 17.92 units
States represented	11
Cities represented	16
Categories represented	4

- **EXPLANATION:**

1. Usable Sales Records

KPI Value: 67 records

After removing the three repeated header rows from the 70 physical rows in the extracted CSV file, **67 usable sales records** remain for

analysis. These records form the primary dataset used for creating the graphs and performing descriptive analysis.

2. Total Recorded Quantity

KPI Value: 95.45 units

The total recorded quantity in the dataset is approximately **95.45 units**. This represents the combined quantity values recorded across the usable sales records. It provides an overall indication of the sales volume represented in the selected sample.

3. Total Recorded Sales

KPI Value: ₹35,448.40

The dataset contains approximately **₹35,448.40 in total recorded sales** across the records where a valid sales amount is available. This value represents the overall revenue captured by the selected sample and is used as the basis for several revenue-related analyses.

4. Average Sales Amount

KPI Value: ₹537.10

The average recorded sales amount is approximately **₹537.10 per valid sales record**. This value provides an indication of the typical sales amount represented by an individual record in the selected dataset.

5. Top-Selling Product

KPI Value: JNE3405 — 9.98 units

JNE3405 is the top-selling product in the selected dataset, with approximately **9.98 units recorded** and it belongs to the category of **Kurta**. This makes it the product with the highest recorded quantity and indicates comparatively strong demand for this product within the sample.

6. Top State by Sales

KPI Value: Maharashtra — ₹9,191.04

Maharashtra is the highest-performing state in terms of recorded sales, generating approximately **₹9,191.04**. It contributes approximately **25.9% of the total recorded sales**, making it the most significant state in the selected dataset.

7. Top City by Sales

KPI Value: Mumbai — ₹5,801.73

Mumbai is the leading city in terms of recorded sales, generating approximately **₹5,801.73**. It contributes around **16.4% of the total recorded sales**, indicating that Mumbai represents an important sales location within the selected sample.

8. Top Revenue Category

KPI Value: Kurta — ₹19,020.51

Kurta is the highest revenue-generating category, with approximately **₹19,020.51 in recorded sales**. It contributes approximately **53.7% of the total sales amount**, making it the dominant category in terms of revenue.

9. Most Common Order Status

KPI Value: Shipped — 37 records (55.2%)

Shipped is the most frequently occurring order status, accounting for **37 out of the 67 usable records**, or approximately **55.2%**. This indicates that more than half of the records in the selected sample are associated with the shipped status.

10. Highest-Demand Size

KPI Value: 3XL — 17.92 units

The **3XL size** records the highest quantity sold, with approximately **17.92 units**. XS follows closely with approximately 17.82 units. This shows that 3XL has the highest recorded size-level demand within the selected sample.

11. Number of States Represented

KPI Value: 11 states

The dataset contains sales records associated with **11 different states**. This indicates that although the dataset is relatively small, it covers multiple geographical regions and allows regional demand to be examined.

12. Number of Cities Represented

KPI Value: 16 cities

The selected dataset contains records from **16 different cities**. This provides a city-level geographical perspective and allows the analysis to

identify locations such as Mumbai that contribute relatively higher sales amounts.

13. Number of Product Categories

KPI Value: 4 categories

The dataset contains **four major product categories: Kurta, Set, Western Dress, and Top**. These categories allow the project to compare both revenue and quantity across different types of products and identify the strongest-performing category.

15. Next Steps to Move from Descriptive → Predictive **From Descriptive Analysis to Predictive Analysis**

The next step would be to extend this analysis toward **predictive analytics**, where historical data can be used to estimate future outcomes and identify potential business risks.

1. Demand Forecasting by Product and Region

The first major predictive step would be to develop a **demand forecasting model for individual products and geographical regions**.

Historical information such as product, category, quantity sold, state, city, date, and fulfilment method could be used to estimate future demand. For example, if a product consistently shows increasing demand over several weeks, a predictive model could estimate its expected demand for the following period.

This could help the business determine:

- Which products are likely to sell more?
- Which regions may experience increased demand?
- How much inventory should be maintained?
- Which products may require earlier replenishment?

2. Predicting Inventory Requirements

The second predictive application would be **inventory requirement prediction**. The descriptive analysis already identifies products and categories with relatively high recorded demand. For example, **Kurta is the leading category in both recorded revenue and quantity**, while JNE3405 is the highest-quantity product in the selected sample.

A predictive system could identify products that are likely to experience high demand and recommend maintaining sufficient stock.

The objective would be to reduce two common inventory problems:

High demand + insufficient stock → Stock-out risk

Low demand + excessive stock → Excess inventory

A future system could therefore move from simply showing current sales to estimating **future demand and expected inventory requirements**.

3. Anomaly Detection

Another useful predictive application would be **anomaly detection**.

Instead of only displaying the current order-status distribution, sales amount, or quantity sold, a predictive system could automatically identify unusual changes in the data.

For example, it could detect:

- An unusually large increase in product sales
- A sudden fall in sales
- An unexpected increase in cancelled orders
- An unusual change in regional demand
- A sudden decrease in product ratings
- An unusual fulfilment pattern

For example, if a product normally sells 20–30 units per day but suddenly records 100 units, the system could flag this as an **unusual demand spike** requiring further investigation.

4. Predicting Order Cancellation

The current report describes the **Order Status Distribution**, where Shipped is the most common status and Cancelled represents a smaller portion of the selected records.

With a larger historical dataset, this information could be extended into a **cancellation prediction model**.

The model could use variables such as:

Product category, Product size, Sales amount, State, City, Previous order behaviour to estimate whether a future order has a higher probability of being cancelled.

This would change the analysis from simply answering:

“How many orders were cancelled?”

to:

“Which types of orders are more likely to be cancelled?”

That would provide more actionable information for business decision-making.

5. Regional Demand Prediction

The current report identifies **Maharashtra as the highest-sales state** and **Mumbai as the highest-sales city** in the selected sample.

With more historical records, regional data could be used to predict future demand for different locations.

For example, a model could estimate:

“Which states are likely to generate the highest sales next month?”

This could support:

- Regional inventory allocation
- Warehouse planning
- Delivery capacity planning
- Regional promotions
- Marketing decisions

Instead of simply reporting where sales **already occurred**, predictive analytics would estimate where demand **may occur in the future**.

6. Product Performance Prediction

The current analysis identifies the **Top 10 Best-Selling Products** based on quantity.

This can be extended into a predictive model that estimates which products are likely to become high performers in the future.

For example:

~Current descriptive analysis:

JNE3405 has the highest recorded quantity in the selected sample.

~Future predictive analysis:

Based on historical sales patterns, product characteristics, category performance, and regional demand, which products are likely to become the next high-demand products?

This would help businesses identify potential **high-growth or “hero” products** before demand reaches its peak.

7. Customer and Purchase Behaviour Analysis

If customer-level information were available in a larger dataset, the next stage could involve studying **repeat purchasing and customer behaviour**.

For example, the analysis could determine:

- Which customers purchase repeatedly?
- Which categories are frequently purchased together?
- How often do customers return?
- Which products encourage repeat purchases?
- Which customer groups generate the most revenue?
-

17. Appendix: Glossary

Order Status — The current stage or condition of an order, such as **Shipped, Cancelled, or Shipped – Delivered to Buyer**.

Sales Amount — The monetary value associated with a sales transaction, used to calculate total and category-wise sales.

Quantity — The number of units recorded or sold for a particular order or product.

Category — The product group to which an item belongs, such as **Kurta, Set, Western Dress, or Top**.

Regional Demand — The level of sales or quantity recorded across different geographical locations such as states and cities.

KPI — Key Performance Indicator; a measurable value used to evaluate important aspects of sales performance, such as total sales, quantity sold, top-selling product, and leading state.

Descriptive Analysis — The process of summarizing historical data using numerical measures, KPIs, tables, and graphs to understand existing sales patterns.

AOV — Average Order Value; the average sales amount generated per order, calculated as **Total Sales ÷ Number of Valid Orders**.